higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

PROCESS TECHNOLOGY
NQF LEVEL 3

(6050353)

12 November 2019 (X-Paper)
09:00–12:00

This question paper consists of 6 pages.

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TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Start each section on a NEW page.
5. Use only BLUE or BLACK ink.
6. Write neatly and legibly.

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SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.1–1.5) in the ANSWER BOOK.

1.1 A mixture of TWO or more substances that have the same composition is …

A heterogenous.
B homogenous.
C continuous.
D stable.

1.2 The purpose of a chemical reaction is to …

A extract a solvent
B promote mixing
C form a completely new product
D produce different chemicals

1.3 … is not ONE of the sections of the MSDS.

A Chemical identity
B Firefighting measures
C Health hazards
D Biological solids

1.4 Heat exchangers are widely used in …

A chemical plants.
B pharmaceutical industries.
C hydromining.
D biological solids.

1.5 When a fluid flows at a 90° angle with the other, the flow is …

A vertical.
B horizontal.
C perpendicular.
D circular.

(5 × 1) **[5]**

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 Volume percentage of a liquid is another type of concentration.

2.2 Density is a measure of mass per unit area.

2.3 In a parallel flow, both fluids enter at different ends and flow in different directions.

2.4 Precipitation fouling is also known as crystallisation fouling.

2.5 A process flow diagram is a diagram used in engineering to indicate the general flow of plant process and equipment.

(5 × 1) **[5]**

QUESTION 3

Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (3.1–3.5) in the ANSWER BOOK.

3.1 A process where a liquid is removed from products in a solid, liquid or gas phase is called …

3.2 Dryers that use drums for drying products in slurry, liquid or paste form are known as … dryers.

3.3 … reduces the level of humidity in the air.

3.4 Silica gel is a/an … used to remove moisture from gases.

3.5 A/An … is a positively charged ion.

(5 × 1) **[5]**

TOTAL SECTION A: 15

SECTION B

QUESTION 4

4.1 A material safety data sheet is a detailed document that helps to select safe chemical products, indicates how to handle them safely and to act correctly in an emergency.

Name FIVE sections of MSDS and explain each. (5 × 2) (10)

4.2 List FIVE factors that influence the mixing process. (5)

4.3 Provide names of the heat exchangers below:

4.3.1 Heat exchanger used in large chemical plants

4.3.2 Heat exchanger used for heat exchange operations where both the shell and the tube side are dirty

4.3.3 Heat exchanger made of thin plates with corrugated flow channels held together by a frame

4.3.4 It reclaims heat energy from a heating ventilation and air-conditioning system or industrial processes.

(4 × 1) (4)

4.4 List FOUR disadvantages of forced draught air coolers. (4)

4.5 Name TWO major components of a cooling system. (2)

[25]

QUESTION 5

5.1 Name and describe basic configurations for the direction of the flow of air in relation to the falling water through the packing in the cooling towers. (4)

5.2 Write brief notes on the following components of a cooling tower:

5.2.1 Distributors (6)

5.2.2 Spray nozzles (5)

5.3 Write a balanced chemical reaction for the oxidation of methane. (5)

5.4 List FIVE applications of paddle dryers as used in industries. (5)

5.5 Why are the heated drums of the drum dryers made up of cast iron, gun metal and stainless steel? (2)

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5.6 A hangup is a problem that normally occurs in spray dryers.

Name THREE indicators for this problem in this type of dryer. (3)
[30]

QUESTION 6

6.1 Name THREE types of desiccants used in industries. (3)

6.2 Name the THREE types of twin tower dessicant dryers that are available. (3)

6.3 Thermal swing is another type of adsorption process.

Give TWO of its uses. (2)

6.4 Define the term *ion exchange resins.* (2)

6.5 Describe the basic working process of ion exchange. (5)

6.6 Write brief notes on the following:

6.6.1 Synthetic organic ionic exchangers

6.6.2 Colloidal particles

(2 × 5) (10)

6.7 List FIVE problems that may cause poor performance of coagulation, flocculation and flotation separation processes in water purification. (5)
[30]

TOTAL SECTION B: 85
GRAND TOTAL: 100

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